

iLLD-TC2-v1.21.0

Release Notes

Product name: iLLD

Release number and version: iLLD-TC2-v1.21.0

Type of release: Alpha*

Release method: Release Manager, Github

Processor platform: TC22xA, TC23xA, TC26xB, TC27xD, TC29xB

Date: 2025-11-07

Previous release number and version: 1.20.0

About this document

Scope and purpose

This document details the release contents, all known issues in the release, and the changes from the last release, together with information on tools and support changes.

New issues identified since the last release of this document are detailed first, followed by all issues identified in previous versions of this release.

*Note: * This release is not intended for production use.*

Note: All source files contained in this release package are subject to Boost Software License - Version 1.0 - August 17th, 2003 - as mentioned in the Disclaimer Header of each file

Attention: **Refer to the Limitations and Deviations chapter before using the software for integration.**

Intended audience

This document is intended for anyone using the iLLD software.

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1 Release contents

1.1 Release overview

This release is of Alpha quality.

1.2 Released items

This release contains the `iLLD-TC2-v1.21.0_Drivers_And_Demos.zip` file.

Table 1 Release zip contents

Package content	Description
<code>iLLD-TC2-v1.21.0.exe</code>	Installer for A1G Source, documentation and Demos
<code>iLLD-TC2-v1.21.0-Releasenotes.pdf</code>	Aurix 1G iLLD Release Notes
<code>BaseProjects_TC2xx_V1_21_0.exe</code>	TC2xx Base Project Installer
<code>BaseProjects_TC2xx_V1_21_0_ReleaseNotes.pdf</code>	Base Project Installer Release Notes

The installer exe "`iLLD-TC2-v1.21.0.exe`" contains the following folders and files mentioned in Table 2

Table 2 Installer content

Package content	Description
<code>TC22A-v1.21.0</code>	Source and documentation files for TC22xA derivative
<code>TC23A-v1.21.0</code>	Source and documentation files for TC23xA derivative
<code>TC26B-v1.21.0</code>	Source and documentation files for TC26xB derivative
<code>TC27D-v1.21.0</code>	Source and documentation files for TC27xD derivative
<code>TC29B-v1.21.0</code>	Source and documentation files for TC29xB derivative
<code>iLLD-TC2-v1.21.0-Releasenotes.pdf</code>	Aurix 1G iLLD Release Notes
<code>Demos-v1.21.0</code>	Demo files

Note:

1. User manual for each device will be available at location `iLLD-TC2*-v1.21.0\Doc`
2. The TC21x shall use the drivers and template projects of TC22x
3. The License text for iLLD has been updated

1.3 Compatibility

1.3.1 Modifications in DMA

The DMA `IfxDma_isChannelReset` API has been updated since it had a bug. It was returning FALSE if channel was RESET and TRUE if channel is not RESET. This has been fixed now, and user will need to use the API appropriately.

2 Tool information

The tool information is provided in the following table.

Table 3 Tool details

Tool description	Version details
Compiler	Tasking v6.3r1 Gnu v4.9.3.0 Windriver: v5.9.6.6 GCC 11.3.1 20221230
Processor platform	TC22xA, TC23xA, TC26xB, TC27xD, TC29xB
Evaluation hardware	TC2xx: Target hardware (Triboard)

3 Summary of changes

This chapter describes the fixes for issues from previous version(s).

3.1 Issues fixed in release V1.21.0

Issue Type	Summary	Description
Enhancement	Return value of IfxStm_getTicksFromMilliseconds and IfxStm_getTicksFromMicroseconds	[2356] Return type of IfxStm_getTicksFromMicroseconds and IfxStm_getTicksFromMilliseconds changed from sint32 to uint32
Bug	IFX-250214-1603954] Potential iLLD error in setting VADC_SYNCTR Register	[2702] Corrected the setting of EVALRy bitfields of the SYNCTR register, for both master and slaves.

3.2 Issues fixed in release V1.20.0

Issue Type	Summary	Description
Enhancement	[TC2xx] [ASCLIN][QSPI]: Synching A2G changes to A1G	[2589] Dependency on StdIf layer removed from ASCLIN

3.3 Issues fixed in release V1.19.0

Issue Type	Summary	Description
Bug	TC2xx ASSW: C++ constructor causing issue for Gnuc & Gcc compiler	[2376] A trap is observed from Ifx_C_Init method in which __init is being called for cpp initialization, linker needs to update correctly.
Enhancement	Remove the motor control SW specific types from Ifx_Types.h	[2103] Moved the application specific enum/struct defined in ifx_types.h to IP or common file.
Enhancement	ASCLIN baud rate does not change properly	[2426] ASCLIN module's baud rate algorithm updated to support lower baud rates
Enhancement	[TC2xx][TC3xx][ASCLIN]: The function IfxAsclin_setBitTiming() calculates incorrect clock divider settings	[2580] The algorithm to calculate clock divider is improved in function IfxAsclin_setBitTiming()

3.4 Issues fixed in release V1.0.1.18.0

Issue Type	Key	Summary
Enhancement	MCMETILLD-2283	[ISOLAR-57187] SAK_TC366DP_64F300S_AA - CCU6 iLLD discrepancy
Enhancement	MCMETILLD-2318	Installer for iLLD
Enhancement	MCMETILLD-2322	TC2x & TC3x PSIS: Provide bypass for GCR.CENx bit in initChannel and add api to enable channel

Summary of changes

Enhancement	MCMETILLD-2330	TC2x & TC3x PSIS: Fix for unintended removal of channelTrigger config in GCR register
Enhancement	MCMETILLD-2085	ASSW: Use of uninitialized global variable in CPU sync-event-wait mechanism, in slave cores
Bug	MCMETILLD-2183	ASSW: The default value of PINDIS in BMI of boot mode headers should be disabled
Bug	MCMETILLD-2309	Bugs in the intrinsics - saturate functions
Bug	MCMETILLD-2314	[ISOLAR-59384] Wrong _START4 for Core 4
Bug	MCMETILLD-2315	IfxCpu_Trap_contextManagementError has rslcx instruction while the lower context is not saved.
Bug	MCMETILLD-2334	Fix gcc compiled elf flashing in ude
Enhancement	MCMETILLD-2055	Add compiler specific define to declare weak a function definition
Enhancement	MCMETILLD-2348	Optimization needed in selecting the fixed clock Frequency for GTM CMU module

3.5 Issues fixed in release V1.0.1.17.0

Issue Type	Key	Summary
Bug	MCMETILLD-2297	iLLD Egtm PWM, IfxEgtm_Pwm_disableChannelsSyncUpdate defect
Bug	MCMETILLD-2298	iLLD Egtm PWM, startSyncedGroups function defect
Enhancement	MCMETILLD-2280	ASSW and Template Project needs to be updated to support C++ constructors and destructors
Bug	MCMETILLD-2282	IfxAsclin_Lin.c wrong initialization of variable flagsenable
Enhancement	MCMETILLD-2285	GTM PWM: Non consecutive (A)TOM channels in a PWM group
Enhancement	MCMETILLD-2286	0000061474-4474: Enhancements for SENT SPC
Bug	MCMETILLD-2292	IfxGtm_Tom_Ch_setCompareZeroShadow() stores 32-bit value into 16-bit register
Bug	MCMETILLD-2293	Fix gcc compiler warnings for I2C
Enhancement	MCMETILLD-2299	Assw: Place the software_init_hook after C init
Bug	MCMETILLD-2304	PWM Interrupt Handler doesn't work in all the use cases
Enhancement	MCMETILLD-2310	Backport from A3G to A2G for legacy COM modules
Enhancement	MCMETILLD-2075	Mistakes within documentation
Enhancement	MCMETILLD-2240	[ISOLAR-55073] GCC support in iLLD

3.6 Issues fixed in release V1.0.1.16.0

Issue Type	Key	Summary
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Summary of changes

Bug	MCMETILLD-2225	[ISOLAR-50124] iLLD IfxFlash_loadPage2X32 Missing DSYNC
Enhancement	MCMETILLD-2114	implicit conversion from 'float32' to 'double'
Enhancement	MCMETILLD-2152	TC23x: iLLD: Request to add DTM support in Gtm_Tom_PwmHl drivers
Enhancement	MCMETILLD-2178	[ISOLAR-44730] FlexRay - Encoding for additional CMD and POC state values
Bug	MCMETILLD-2182	[ISOLAR-44827] GTM ATOM PWM iLLD driver behaviour
Enhancement	MCMETILLD-2210	QSPI Delay Params, backport from TC4X
Enhancement	MCMETILLD-2230	Reverse Port IfxGtm_Pwm driver from TC4x
Bug	MCMETILLD-2224	IFXGTM DTM: Wrong parameter type
Bug	MCMETILLD-2237	(E)GTM PWM Update Duty/Phase Immediate would not work in corner cases
Enhancement	MCMETILLD-2094	Gtp12IncEnc is reversed by default
Enhancement	MCMETILLD-2170	[ISOLAR-41639] GTM TIM TDU Timeout - Enhancement Request
Enhancement	MCMETILLD-2242	Update read and write APIs. Read: Move IfxI2c_clearAllDtrInterruptSources after polling for the events in while loop. Write: Wait for event after FIFO write. Add while (!(i2c->RIS.U)).
Bug	MCMETILLD-1417	iLLD CCU6 driver `IfxCcu6_PwmBc_initModule()` bug report

3.7 Issues fixed in release V1.0.1.15.0

Issue Type	Key	Summary
Enhancement	MCMETILLD-2084	Incorrect implementation of IfxMultican_Can_MsgObj_init regarding the FIFO structure
Enhancement	MCMETILLD-2118	OSS compliance: Legal recommended header format missing in file AsclnShellInterface.h
Bug	MCMETILLD-2159	Missing analog input variables (IfxVadc_Vadcg_In) for the TC22x and TC23x devices in the _pinmap files
Enhancement	MCMETILLD-2173	ALM0[18] and ALM3[27] missing in SMU alarm list
Bug	MCMETILLD-2179	[ISOLAR-44727] FlexRay - Wrong default configuration for MacroTickInitialOffset value for channel B
Bug	MCMETILLD-2182	[ISOLAR-44827] GTM ATOM PWM iLLD driver behaviour
Bug	MCMETILLD-2188	GTM: A1G derivatives missing IfxGtm_Tim_Ch_setShadowCounter API
Bug	MCMETILLD-2197	[ISOLAR-46818] IfxScuWdt_initSafetyWatchdog CLRIRF Logic Inverted
Bug	MCMETILLD-2209	FCE: A wait of minimum 2 clock cycles are needed after the input is written into IR register

3.8 Issues fixed in release V1.0.1.14.0

Issue Type	Key	Summary
Enhancement	MCMETILLD-2177	[ISOLAR-44140] CCU6: Bug in calculation of prescaler
Bug	MCMETILLD-2172	[ISOLAR-42223] GTM CMU Fixed Clock Frequency Calculation
Bug	MCMETILLD-2171	[ISOLAR-42337] GTM (A)TOM Force Update Issue
Bug	MCMETILLD-2169	[ISOLAR-41931] Bug in CRC32 tasking ILLD implementation
Bug	MCMETILLD-2167	iLLD setting CCUCON1 bug
Bug	MCMETILLD-2166	IfxIom_getFrequency function returns double the IOM frequency
Enhancement	MCMETILLD-2151	Wrong implementation of "IfxEray_setNewDataInterruptDestination" in ERAY driver
Bug	MCMETILLD-2093	IfxCcu6_TimerWithTrigger_updateInputFrequency() function does not take into consideration the TCTR0.B.T12PRE bit for calculating the prescaler value

3.9 Issues fixed in release V1.0.1.13.0

Issue Type	Key	Summary
Bug	MCMETILLD-2148	AxG: CSA: Possible dereference through NULL pointer in iLLD start-up code
Bug	MCMETILLD-2147	I2C read wrong flags are checked
Bug	MCMETILLD-2136	TC2XX CAN incomplete Gateway configuration, TOP and BOT configuration missing
Enhancement	MCMETILLD-2088	Incorrect configuration of receive message object in IfxMultican_Can_MsgObj_init() function
Bug	MCMETILLD-2155	Incorrect size of the ERAY configuration parameters
Enhancement	MCMETILLD-2142	Comments and function declaration of initTime function to be removed from Bsp.h

3.10 Issues fixed in release V1.0.1.12.0

Issue Type	Key	Summary
Bug	MCMETILLD-2099	IfxStm_enableOcdsSuspend traps when the debugger is removed
Bug	MCMETILLD-2134	Wrong value for GLOBALCON.RESETS bitfield in QSPI driver
Enhancement	MCMETILLD-2120	MulticanFd demo has configurations mixed for sync jump width and sample point
Enhancement	MCMETILLD-2100	Move SysSe folder out of the iLLD package
Enhancement	MCMETILLD-2076	Headers to be added and adjusted in terms of consistency
Bug	MCMETILLD-2073	Incomplete asm instruction name
Bug	MCMETILLD-2092	Intrinsics: Outdated implementation of CRC32-related functions

Summary of changes

Bug	MCMETILLD-2101	Intrinsic function getA11 does not comply with EABI manual
Enhancement	MCMETILLD-2074	Improvement in macro definition for IFX_INLINE
Enhancement	MCMETILLD-2059	Update license text
Bug	MCMETILLD-2036	Infinite while loop due to cleared TXEND flag

3.11 Issues fixed in release V1.0.1.11.0

Issue Type	Key	Summary
Enhancement	MCMETILLD-2023	Add getCodeFromDataLength(), getDataLengthFromCode()
Enhancement	MCMETILLD-1969	Wrong location for define ENDL
Enhancement	MCMETILLD-1964	Potential paranthesis issue in i2c driver
Enhancement	MCMETILLD-1962	SPI default values in half-duplex mode cause error in MMIC communication
Bug	MCMETILLD-1961	TIM getDutyPercent does not return DutyPercent
Enhancement	MCMETILLD-1949	Not resolveable compiler warnings visible in release note sheet-A1G
Bug	MCMETILLD-1907	IfxQspi_SpiMaster_write() may lead to TX FIFO overflow
Bug	MCMETILLD-1906	IfxQspi_SpiMaster_write(): The interrupt disable could be reduced
Bug	MCMETILLD-1771	TC27x and TC23x violated VADC master slave initialization sequence
Enhancement	MCMETILLD-615	ASCLIN: LIN interrupts
Enhancement	MCMETILLD-2027	Removal of compiler warnings

3.12 Issues fixed in release V1.0.1.10.0

Issue Type	Key	Summary
Enhancement	MCMETILLD-1858	Rework QSPI coefficients calculation in IfxQspi_calculateExtendedConfigurationValue()
Bug	MCMETILLD-1847	GPT12 encoder driver bug in the speed calculation
Enhancement	MCMETILLD-1819	FIFO is not handled properly
Bug	MCMETILLD-1814	IfxDsadc_Dsadc_initChannelConfig() is not explicitly initializing all members
Enhancement	MCMETILLD-1781	IOM module driver are missing for some derivatives
Bug	MCMETILLD-1777	Ethernet driver not usable with MII
Enhancement	MCMETILLD-1702	iLLD configuration
Enhancement	MCMETILLD-1701	errors and new function in iLLD A1G MultiCAN
Enhancement	MCMETILLD-1541	Add support for Ethernet MII-lite pinning
Enhancement	MCMETILLD-1181	pointer to MDC and MDIO pins is missing in MiiPins structure

Summary of changes

Enhancement	MCMETILLD-946	SCU: Use same IfxScuWdt_calculateLfsr accross all Toolchains
Enhancement	MCMETILLD-582	ETH: IfxEth_Phy_Pef7071 should be able to change configuration for LEDs
Enhancement	MCMETILLD-581	I2C: problem with Windriver 5.9.2 compiler/optimisation

3.13 Issues fixed in release V1.0.1.9.0

Issue Type	Key	Summary
Bug	MCMETILLD-1822	Mtu: TC2xx : wrong value assigned to reserved bit in register MCONTROL causing field failures
Bug	MCMETILLD-1768	TC26xB : Compilation fail in IfxGtm_Trig.c with Tasking compiler
Bug	MCMETILLD-1753	incorrect use of __asm keyword in IfxCpu_IntrinsicsTasking.h
Enhancement	MCMETILLD-1752	C++ support: __asm is not supported by TASKING C++ compiler
Enhancement	MCMETILLD-1751	C++ support: Tasking g++ option for C++ is not supported
Bug	MCMETILLD-1732	IfxCcu6_enableAdditionalPrescaler don't enable Additional Prescaler
Enhancement	MCMETILLD-1713	Wrong Data type in IfxAsclin_Asc_isrReceive function
Enhancement	MCMETILLD-1668	CCU6: ICU iLLD resolution enhancement to 16-bit
Bug	MCMETILLD-1635	The QSPI Error interrupt is not initialized when DMA is used
Bug	MCMETILLD-1604	IfxAsclin_Asc_write() hangs up if count > fifo size
Enhancement	MCMETILLD-1562	IfxAsclin_write: usage of const in front of pointer void *data.
Enhancement	MCMETILLD-1317	CCU6_ICU : issue setting the frequency
Enhancement	MCMETILLD-1269	Make intrinsic independent from compiler vendors
Enhancement	MCMETILLD-446	MTU: Implement errata workarounds into IfxMtu driver
Enhancement	MCMETILLD-407	CPU : IFX_VECTOR_CPU* are not documented

3.14 Issues fixed in release V1.0.1.8.0

Issue Type	Key	Summary
Enhancement	MCMETILLD-1722	Implicit declaration of function IfxScuCcu_getMscFrequency
Bug	MCMETILLD-1708	IfxQspi_SpiMaster.maximumBaudrate is not assigned but used for some tests
Bug	MCMETILLD-1707	Handle->dma.useDma is not initialized by IfxQspi_SpiMaster_initModule() when DMA is not used
Enhancement	MCMETILLD-1632	Dma Demo: Data alignment issue with Tasking compiler

3.15 Issues fixed in release V1.0.1.7.0

Issue Type	Key	Summary
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Summary of changes

Enhancement	MCMETILLD-1641	CRC32 intrinsic to be added for DCC
Enhancement	MCMETILLD-1610	cstart: reserving 2 CSAs beyond LCX
Bug	MCMETILLD-1354	GPT12 encoder driver `IfxGpt12_IncrEnc.c` bug report
Bug	MCMETILLD-1588	Lin: issue with setDataLength api
Bug	MCMETILLD-1621	Intrinsic function for __cmpAndSwap with DCC not working

3.16 Issues fixed in release V1.0.1.6.0

Issue Type	Key	Summary
New Feature	MCMETILLD-1544	MSC Issues regarding BaudRate calculation APIs and ABRA status flags
Enhancement	MCMETILLD-1545	DSADC documentation mentions usage of IfxDsadc_Rdc_init() before IfxDsadc_Rdc_initConfig()

3.17 Issues fixed in release V1.0.1.5.0

Issue Type	Key	Summary
Bug	MCMETILLD-1515	Missing ATOM IRQs in iLLD of TC29xB
Bug	MCMETILLD-867	SFR : Wrong register files for TC27x D-Step
Bug	MCMETILLD-1283	All: CPU: Incorrect documentation for Trap usage

3.18 Issues fixed in release V1.0.1.4.0

Issue Type	Key	Summary
Enhancement	MCMETILLD-1442	Update the Ifx_Typesreg.h file with the comments regarding access possibilities for memories
Enhancement	MCMETILLD-1355	IfxScuCcu driver has a rounding problem
Bug	MCMETILLD-1356	AURIX iLLD for module clock configuration

3.19 Issues fixed in release V1.0.1.3.0

Issue Type	Key	Summary
Bug	MCMETILLD-1405	TC39x A-step Port regdef file wrong PDR0 bit-field definitions
Enhancement	MCMETILLD-1404	Aurix2G Startup: For TC38x to enable LBIST execution along with LBISTREQ bit LBISTREQRD bit has to be set
Bug	MCMETILLD-1400	Asclin ASC/UART: Not able to set baudrates below 13000
Enhancement	MCMETILLD-1392	PHY configuration write, read APIs is required.
Bug	MCMETILLD-1391	In API IfxGeth_Eth_sendTransmitBuffer, the bit TDES3.R.CIC_TPL must be set to 0x3
Bug	MCMETILLD-1390	GETH GPCTL.B.ALTIO for MDIO functionality is wrong
New Feature	MCMETILLD-1363	TC3xx: Startup address to be updated to 0xA00A0000

Summary of changes

Bug	MCMETILLD-1347	EMEM writes not working
Bug	MCMETILLD-1346	AsclnShellInterface demo is not working in TC38x
Modification	MCMETILLD-1324	Change of member name calibmode in IfxEmem_TileConfigMode
Bug	MCMETILLD-1323	Only-internally-used register information is contained in the reg files of TC39A
New Feature	MCMETILLD-1320	DMA : New API's needed to get the SDCRC , RDCRC values, To enable SWAP Byte, To switch resource partitions and to return DMA ERR SRC pointer.
Bug	MCMETILLD-1316	issue in iLLD function IfxCan_Node_enableConfigurationChange
Bug	MCMETILLD-1315	wrong register poll in IfxI2c_I2c_read
Bug	MCMETILLD-1303	SSW cfg of democode not runnable on Triboard
Bug	MCMETILLD-1302	ASCLIN oversampling factor setting conflict
Bug	MCMETILLD-1298	IfxMtu_enableModule() does not care for Cpu Endinit
Improvement	MCMETILLD-1279	EMEM: Unlock sequence
Improvement	MCMETILLD-1266	All: CPU: Replace Gnu style asm macro with windriver style for Diab compilation
Enhancement	MCMETILLD-1236	SW to poll MTU_MEMSTAT[x] or IRQ rather than MTU_MEMTEST/MTU_MEMDONE[x] for derivatives with large DSPRs
New Feature	MCMETILLD-1185	TC3xx: PMS/Smu Stdb: MONBIST APIs documentation update
Bug	MCMETILLD-1106	QSPI: Run mode requested for QSPI slave instance before configuring this instance for slave mode
Bug	MCMETILLD-1017	QSPI baudrate target 4MBaud results in 3.3MBaud instead
Bug	MCMETILLD-619	QSPI : IfxQspi_SpiMaster_initChannel configures wrong delays
Enhancement	MCMETILLD-431	QSPI: AURIX: Configuring Data Length in Bytes for QSPI transfer not supported
Enhancement	MCMETILLD-342	QSPI : IfxQspi_SpiMaster_initModule() does not efficiently use IfxDma_Dma_initChannel()
Enhancement	MCMETILLD-1309	Software managed interrupts are not supported for Aurix2G for cores CPU1 to 5

3.20 Issues fixed in release V1.0.1.2.0

Table 4 Issue summary

Issue type	Key	Summary
Enhancement	MCMETILLD-1300	Pinmaps for LFBGA516 and LFBGA292 packages for TC3xx required
Bug	MCMETILLD-1295	IfxGtm_Dtm_setClockSource should not have parameter channel (unused)
Bug	MCMETILLD-1287	IfxCpu_waitEvent timeout is wrong
Bug	MCMETILLD-1285	GTM: Correct Psm enable F2A stream and disable F2A stream functions
Enhancement	MCMETILLD-1281	TC3xx: SCU_CCU: Default clock configurations + validity
Modification	MCMETILLD-1280	Ifx_Ssw_Infra.h: Endint functions can be optimised by reading complete register instead of blocking while()

Summary of changes

Modification	MCMETILLD-1277	Remove the DSPR0 check functions in MTU files
Bug	MCMETILLD-1276	Add #pragma optimise macros for GNU and Tasking in startup files
Bug	MCMETILLD-1275	IfxHspdm can't be compiled with non-internal sources
Improvement	MCMETILLD-1273	GTM: PSM set channel mode accepts mismatched argument type for channel mode
Modification	MCMETILLD-1271	Correct DTS formula
Bug	MCMETILLD-1268	SDMMC adma2 API needs correction on check for transfer complete
Improvement	MCMETILLD-1267	TC3xx: Ssw: Add noadjust directive to setAddressReg macro
Improvement	MCMETILLD-1265	All: PORT: IfxPort_getIndex API shall return IfxPort_Index type instead of sint32
Improvement	MCMETILLD-1264	All: InternalMux, Intrinsics, Mtu, Pms: Remove warnings from Dcc
Bug	MCMETILLD-1263	TC39xx: SSW: Intrinsics macros for DCC are wrongly modified
Bug	MCMETILLD-1262	TC3xx: MTU: Default constants used for gang configurations shall be updated
Improvement	MCMETILLD-1260	TC3xx: MTU: IfxMtu_runMbistConfigRange function shall clear the ECCD errors before test
New Feature	MCMETILLD-1259	TC3xx: Startup Sw: CPU 0 disable shall be made possible
Bug	MCMETILLD-1258	TC3xx: Startup Sw: Ifx_Ssw_Tc1.c does not start CPU3 when CPU2 is disabled
Modification	MCMETILLD-1251	Switch Clock Source to fPLL (CLKSEL=1) before SYS_PLL is locked
Support Issue	MCMETILLD-1242	incorrect definition of Spu configuration slot
Modification	MCMETILLD-1228	IFXCPU_ALLCORE_DONE macro is supporting to configure the value. So the naming convention has to be updated as IFXCPU_CFG_ALLCORE_DONE
Enhancement	MCMETILLD-1209	TC39xA iLLD does not correctly support TX Event FIFO of MCMCAN module
New Feature	MCMETILLD-1187	Add function to enable watchdog with debugger connected
Modification	MCMETILLD-547	CPU: Change return type of IfxCpu_getCoreId() from IfxCpu_ResourceCpu to IfxCpu_Id
Bug	MCMETILLD-1306	All Aurix1G: iLLD files are not compiling with GNUC when -std=c99 option is enabled
Bug	MCMETILLD-1297	All: iLLD files are not compiling with GNUC when -std=c99 option is enabled
Bug	MCMETILLD-1257	All: IfxCpu: IfxCpu_getPerformanceCounter API gives compilation error
Bug	MCMETILLD-1256	TC3xx: MTU: IfxMtu_runMbist result is negative for peripheral RAMs during Startup Software
Bug	MCMETILLD-1130	TC39xA: IfxMtu_clearSram doesn't clear the CPU0 DSPR.
Bug	MCMETILLD-907	SFR : Wrong EVADC register definitions

3.21 Issues fixed in release V1.0.1.1.0

Table 5 Issue summary

Issue type	Key	Summary
Bug	MCMETILLD-1023	Wrong conversion time calculation in VADC
Modification	MCMETILLD-1068	CPU: Documentation of trap hooks needs enhancement
Bug	MCMETILLD-1105	wrong calculation of CAN frequency
Support Issue	MCMETILLD-1170	iLLD startup function IfxCpu_getRandomValue fails to be inlined (e.g. using O3)
Bug	MCMETILLD-1173	__crc32: conflict with Tasking intrinsic and iLLD intrinsic
New Feature	MCMETILLD-1174	__crc32 intrinsic api to be created for GnuC similar to tasking builtin intrinsic
Modification	MCMETILLD-1184	PinMap: remove _Reg/ from all pinMap.h files applicable
Bug	MCMETILLD-1186	Non-destructive inversion test will not be executed when SRAM_CLR is set.
Improvement	MCMETILLD-1188	EDSADC default initialization doesn't include GAINCALx, GAINCTRx, GAINCORR, GLOBRCregisters.
Bug	MCMETILLD-1189	iLLD: OSCVAL is not updated when PLL input deosn't equal fOSCO
Bug	MCMETILLD-1190	Data receive de-synchronization if FIFO fill level is > 1
Bug	MCMETILLD-1192	Bug in IfxIom_Driver_initLam()
Bug	MCMETILLD-1202	wrong while condition in IfxCpu_waitEvent
Bug	MCMETILLD-1215	Gtm: spb clock frequency return value is 0
Bug	MCMETILLD-1230	Wrong data type on RIF iLLD
Enhancement	MCMETILLD-1231	GTM: Clock configuration flow
New Feature	MCMETILLD-730	Disable or enable BMHD definitions from compilation
New Feature	MCMETILLD-785	Aurix 2G: Converter Control Driver
Bug	MCMETILLD-921	IfxDsadc_Dsadc_initRectifier() does not take in account NVALDIS
Bug	MCMETILLD-1014	Wrong function call in float32 IfxGtm_Cmu_getModuleFrequency
Enhancement	MCMETILLD-1040	IfxQspi: NULL_PTR not considered in pin configuration
New Feature	MCMETILLD-1116	index treatment for MODULE_SCU.PMCSR[index].B.REQSLP is wrong in Pms/Std/IfxPmsPm.c
Bug	MCMETILLD-1139	IfxCan: API for global filter configuration missing
New Feature	MCMETILLD-1210	GTM: provide iLLD standard layer for SPE submodule
Improvement	MCMETILLD-782	SCU_CCU: Add update of CCUCONx
Bug	MCMETILLD-1241	All: IfxCpu_emitEvent does not work due to race condition
New Feature	MCMETILLD-1238	TC3xx: MTU default gang configurations
Modification	MCMETILLD-1233	Update mbist ssh configuration with proper default values
Bug	MCMETILLD-1234	Add isync and dsync after Ifx_Ssw_initCSA
Bug	MCMETILLD-1245	TC3xx: PMS EVR setup at startup needs to check EVRSTAT register before configuring
New Feature	MCMETILLD-1250	Add an API in MTU to do MBIST check for all the gangs configured (e.g. IfxMtu_runMbistAll()).

Summary of changes

Bug	MCMETILLD-1229	EVADC wrong calculation of SRC register address
Improvement	MCMETILLD-894	EVADC: provide interface & standard layer functions for fast compare
Modification	MCMETILLD-688	EVADC / EDSADC: Evadc and Edsadc needs meaningful default configuration

3.22 Issues fixed in release V1.0.1.0.0

Table 6 Issue summary

Issue type	Key	Summary
New Feature	MCMETILLD-857	MSC: MSC64 Fast Mode function needs to be created
New Feature	MCMETILLD-830	Intrinsics: Add CRC instruction to the intrinsics
New Feature	MCMETILLD-852	Add popcnt instruction to the intrinsics
New Feature	MCMETILLD-968	MSC: Add MSC64 Command extension(CX) mode
New Feature	MCMETILLD-990	__attribute__((aligned(64))) is used in some of the demo code instead of IFX_ALIGN()
New Feature	MCMETILLD-912	Add new intrinsics for GNUC
New Feature	MCMETILLD-892	QSPI : add error initialization
Enhancement	MCMETILLD-439	SMU: SMU low level driver (Aurix1G)
Enhancement	MCMETILLD-883	Add Suspend API of IPs that support ocds suspend
Enhancement	MCMETILLD-903	ETH: TC39xA - timestamp API's required to test the PTP mode
Improvement	MCMETILLD-1120	TC39x: IFX Interrupts documentation improvement
Improvement	MCMETILLD-904	SCU_CCU: IfxScuCcu_isOscillatorStable() should not fail for 8MHz XTAL in A2G
Improvement	MCMETILLD-950	PSI5S: how to disable channels/channel Trigger Counters in respective Inline functions
Improvement	MCMETILLD-957	PSI5S: Add apis getSuccessfullyReceivedFrameStatus(), stopAscTransactions() and disableAscReceiver()
Modification	MCMETILLD-1067	EDSADC : Define available EDSADC configuration functions public
Modification	MCMETILLD-961	SPU : 32-bit access is needed for PWRSUM register
Bug	MCMETILLD-935	ASCLIN : LIN Function IfxAsclin_Lin_waitForReceivedHeader() waits in infinite loop if header not received
Bug	MCMETILLD-971	Tasking intrinsics __subsu and __addsu does not compile
Bug	MCMETILLD-991	DMA: Typo in IfxLld_Dma_Dma example code
Bug	MCMETILLD-1003	DMA: linked list must be aligned on 32byte / 256bits boundary
Bug	MCMETILLD-1057	Wrong typedef IfxCan_ClockSource
Bug	MCMETILLD-1065	EDSADC: inconsistency in the DICFGx conf Register between ITS and LLD
Bug	MCMETILLD-1114	TC39x:SSW: Wrong const name used in the error message
Bug	MCMETILLD-773	HSSL: TIDADD is written in IfxHssl_Hssl_singleFrameRequest
Bug	MCMETILLD-901	ERAY: kernel reset API when used, goes to trap. observed that the kernel reset is being tried without clearing and setting the cpu & safety endint

3.23 Issues fixed in release V1.0.0.13.0

Table 7 Issue summary

Issue type	Key	Summary
Bug	MCMETILLD-939	Ssw: IfxCpu_CStart0 doesn't enable caches

Summary of changes

New Feature	MCMETILLD-927	SMU : SMU Driver
New Feature	MCMETILLD-926	IOM : IOM driver
New Feature	MCMETILLD-748	DSADC: Added DSADC Resolver to Digital converter (DSADC RDC)
Improvement	MCMETILLD-628	Wrong default clock for TC29x
New Feature	MCMETILLD-911	STM : interface driver
Bug	MCMETILLD-962	GTM : Multi view register in the GTM atom channel is generating trap while accessing then via iLLD API
Bug	MCMETILLD-924	IfxQspi_SpiMaster_getChannelConfig() return wrong value for autoCS
Bug	MCMETILLD-922	IfxQspi_SpiMaster_setChannelBaudrate() baudrate parameter type should be float32 instead of float
Bug	MCMETILLD-921	IfxDsadc_Dsadc_initRectifier() does not take in account NVALDIS
Bug	MCMETILLD-918	QSPI : macros IFXQSPI_BACONSIZ, IFXQSPI_FIFO32BITSIZE, IFXQSPI_GETLONGMODEFIFOSIZE should not be defined in Spilf.h
Bug	MCMETILLD-908	EBU : Wrong pin selection in IfxEbu_Sram_initMemory
Bug	MCMETILLD-837	SCU_CCU : Function "IfxScuCcu_initErayPll" is never returning a fail
Bug	MCMETILLD-866	SCU_CCU : wrong cpu frequency calculation in float32 IfxScuCcu_getCpuFrequency(const IfxCpu_ResourceCpu cpu

3.24 Issues fixed in release V1.0.0.12.0

Table 8 Issue summary

Issue type	Key	Summary
Bug	MCMETILLD-938	IfxSpu shouldn't use read-modify-write accesses
Bug	MCMETILLD-935	ASCLIN : LIN Function IfxAsclin_Lin_waitForReceivedHeader() waits in infinite loop if header not received
Bug	MCMETILLD-931	IfxScuCcu_setSpbFrequency function change SCU_TRAPDIS0 settings
Bug	MCMETILLD-930	HSCT: Register bit CONFIGPHY.PON needs to be set to 1 also during slave interface initialization
Bug	MCMETILLD-915	Enum values IfxCan_ClockSource_asynchronous, IfxCan_ClockSource_synchronous needs to be interchanged
Bug	MCMETILLD-913	CAN iLLD: Incorrect bit timing value calculations for some baudrates
New Feature	MCMETILLD-899	TC39xA: Startup code enhancements
Improvement	MCMETILLD-893	iLLD function in to inline function
Bug	MCMETILLD-891	Bug in Multi Bug CAN driver
Bug	MCMETILLD-876	ASCLIN:LIN function "IfxAsclin_Lin_receiveResponse" has issue
Bug	MCMETILLD-875	ASCLIN: Issue in function IfxAsclin_Lin_receiveHeader.
Enhancement	MCMETILLD-874	MSC init module function need to be updated
Enhancement	MCMETILLD-871	Ethernet functions with enhanced descriptors in ring mode is required.
Improvement	MCMETILLD-864	FCE module interrupt initialization and clearing error flags issue
Improvement	MCMETILLD-859	QSPI: Single, Continuous and Batch mode - fifo mode configurations support to be added
New Feature	MCMETILLD-858	SENT: Watch Dog Timer functionality
Bug	MCMETILLD-846	IfxCan_Can_Baudrate and IfxCan_Can_FastBaudrate contains unused data members

New Feature	MCMETILLD-814	Dma:Channel Suspend Register functions to enable, disable and acknowledge are not present
Improvement	MCMETILLD-604	CAN: Improvements in iLLD function IfxCan_Can_initNode
Bug	MCMETILLD-555	TC22x IfxMultican fails at 1 MBaud

3.25 Issues fixed in release V1.0.0.11.2

Table 9 Issue summary

Summary of changes

Issue type	Key	Summary
Bug	MCMETILLD-854	Additional register definition for MCDS TCZ in IfxMcds_reg.h
Bug	MCMETILLD-849	SPI Reset of State Machine, Rx and TX Fifo
Bug	MCMETILLD-843	xbar register file is buggy
Enhancement	MCMETILLD-840	IfxDts: TC39xA - Dts value to Celsius should be changed
Bug	MCMETILLD-836	wrong structures for bit field GTM_ATOMx_Chx_CTRL
Bug	MCMETILLD-835	Incorrect RABR.OMEM field bit width
Bug	MCMETILLD-834	EBCU Register file has many register missing
Bug	MCMETILLD-831	base address for MCDS SESSID registers is incorrect
New Feature	MCMETILLD-828	EVADC Registers GxTRCTR missing
Enhancement	MCMETILLD-824	default value of SJW and SP in void IfxMultican_Can_Node_initConfig()
Bug	MCMETILLD-822	EVADC: Inconsistency in register bit field file
Enhancement	MCMETILLD-820	IfxCan: Aurix 2G - IfxCan_Can_readMessage should clear Rx buffer flag after data reception.
Enhancement	MCMETILLD-819	IfxCan : Aurix 2G - Add Interface function for TxBufferRequestPending
Enhancement	MCMETILLD-818	IfxCan: Aurix 2G - IfxCan_Can_sendMessage should return status if buffer isn't free
Enhancement	MCMETILLD-815	Remove _Reg/IfxCbs_*
New Feature	MCMETILLD-814	Dma:Channel Suspend Register functions to enable, disable and acknowledge are not present
Enhancement	MCMETILLD-811	Missing IfxFlash_cfg.c in file IfxFlash.xml
Bug	MCMETILLD-808	TC39xA EMEM: EMEM SFR address offset is wrong.
Bug	MCMETILLD-806	Trap handlers corrupt the lower context
Bug	MCMETILLD-801	TC39xA EMEM: EMEM SFR address definition doesn't match ITS
Enhancement	MCMETILLD-796	Missing Overlay Register OVC3_xx, OVC4_xx, OVC5_xx in iLLD - IfxOvc_reg.h
Enhancement	MCMETILLD-793	ASCLIN: LIN Slave Mode in Auto baud enabled needs upper limit and lower limit values calculation function calls
Enhancement	MCMETILLD-792	ASCLIN: iLLD configures LIN slave in autobaud enabled always
Bug	MCMETILLD-780	Trap handler should not use shared global variable
Bug	MCMETILLD-771	IfxCpu_Trap_nonMaskableInterrupt() should not call __debug()
Enhancement	MCMETILLD-770	Trap hooks should be defined explicitly for each trap vector
New Feature	MCMETILLD-762	Cpu:New Feature for CPU Synchronization
Enhancement	MCMETILLD-755	Enhanced Ifx_Shell
New Feature	MCMETILLD-753	Added pulse and shift PWM for GTM ATOM PWMHL driver
New Feature	MCMETILLD-752	Added pulse and shift PWM for GTM TOM PWMHL driver
Enhancement	MCMETILLD-751	Fix CCU6 PWMHL IfxCcu6_PwmHl_setMode() parameter check
Enhancement	MCMETILLD-750	Update GPT21 IncEnc driver
New Feature	MCMETILLD-749	Added GTM TIM input capture feature
Enhancement	MCMETILLD-725	Aurix: ACT Plugin enhancement & plugins update
Enhancement	MCMETILLD-706	Aurix 2G:Enhance IfxPort_setPinPadDriver to support TTL3V3/TTL5V selection
New Feature	MCMETILLD-648	EMEM: Overlay iLLD feature
Enhancement	MCMETILLD-630	TC39x: support for Gtm Edsadc Trig api to configure TIMxINSEL and SAUL,SBLL inputs
New Feature	MCMETILLD-274	Provide IfxSpu (ADAS ADC)

3.26 Issues fixed in release V1.0.0.11.1

Table 10 Issue summary

Issue type	Key	Summary
Enhancement	MCMETILLD-811	Missing IfxFlash_cfg.c in file IfxFlash.xml
Bug	MCMETILLD-808	TC39xA EMEM: EMEM SFR address offset is wrong.
Bug	MCMETILLD-801	TC39xA EMEM: EMEM SFR address definition doesn't match ITS
Improvement	MCMETILLD-755	Enhanced Ifx_Shell
Enhancement	MCMETILLD-751	Fix CCU6 PWMHL IfxCcu6_PwmHl_setMode() parameter check
Enhancement	MCMETILLD-750	Update GPT21 IncEnc driver
Enhancement	MCMETILLD-725	Aurix: ACT Plugin enhancement & plugins update

4 Known issues

This chapter describes the prescribed workarounds for all the open issues identified.

Key	Summary	Impact	Workaround
MCMETILLD-2587	I2C iLLD A1G has set the incorrect value for slave configuration	I2C cannot be configured in slave mode, due to incorrect setting api <code>IfxI2c_configureAsSlave</code>	In <code>IfxI2c_configureAsSlave</code> api, configure the bit <code>MnS</code> to zero.

5 Limitations and Deviations

This chapter describes the limitations and deviations due to software/hardware design constraints.

5.1 Limitations

None

5.2 Deviations

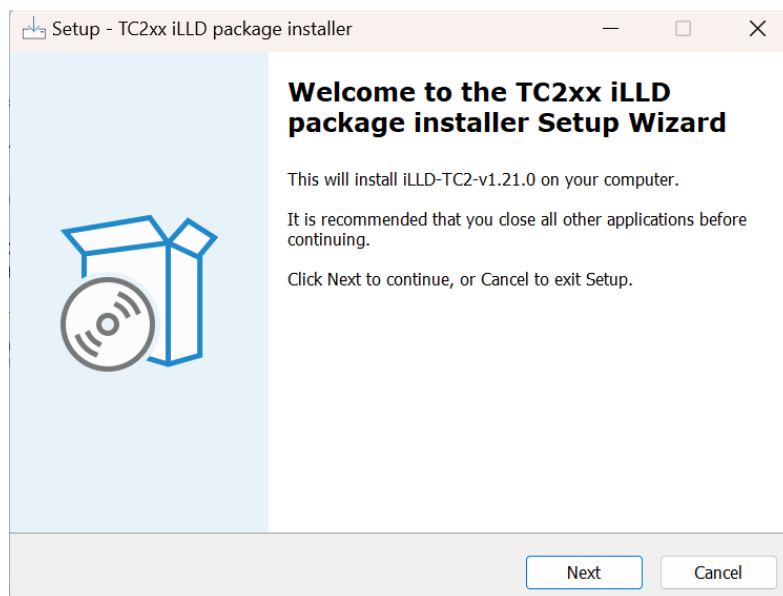
None

6 Installation

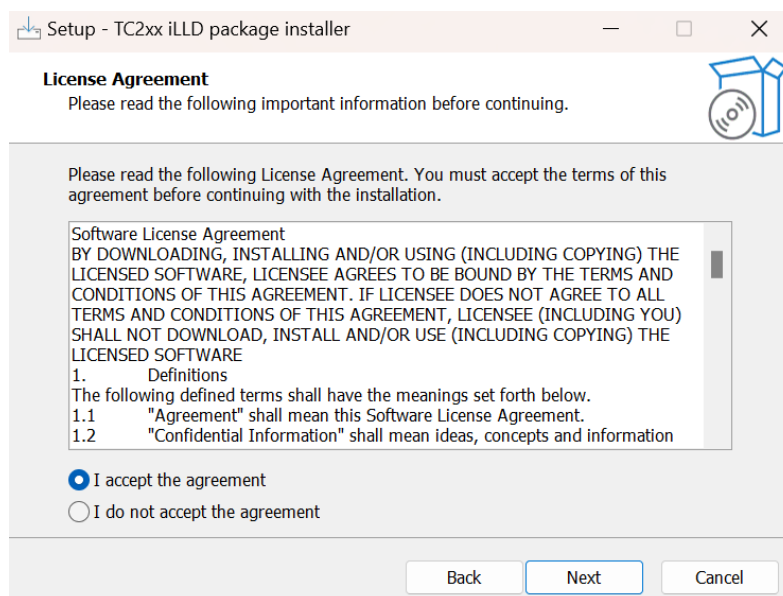
The iLLD software release package is an archive, and it may be extracted to any drive of the computer. There is no specific directory structure needed.

6.1 Package usage

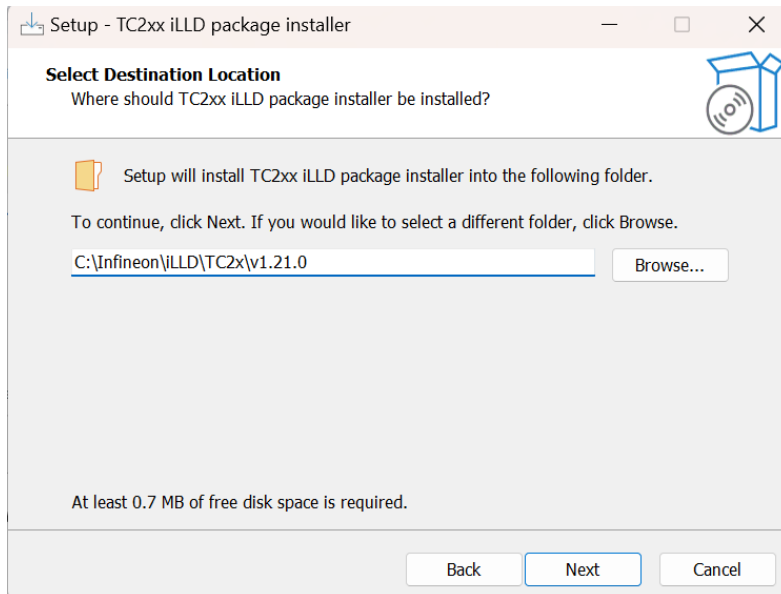
- Copy the iLLD-TC2-<version>_Drivers_And_Demos.zip file into local drive.
- Extract the zip and run the installer file iLLD-TC2-<version>.exe for package installation



- Accept the license and click on “Next”

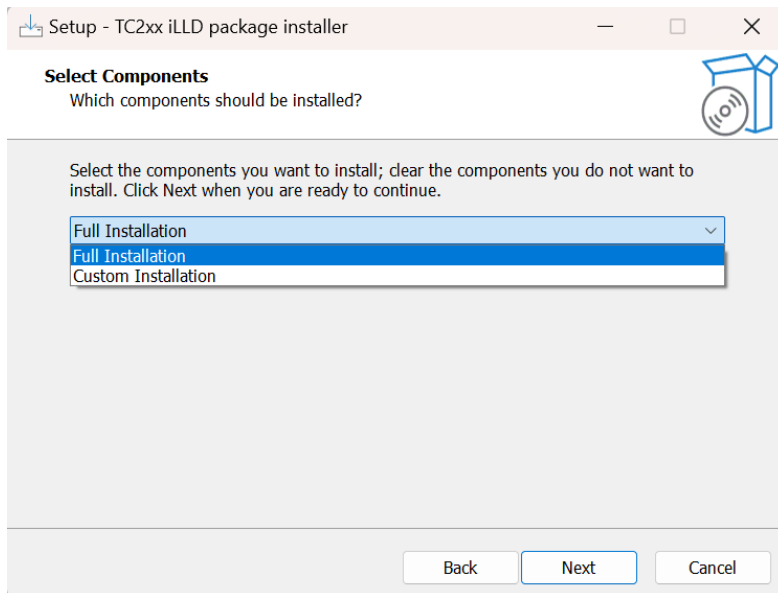


- By default, the package will be installed in `C:\Infineon\iLLD\TC2x\<version>` path



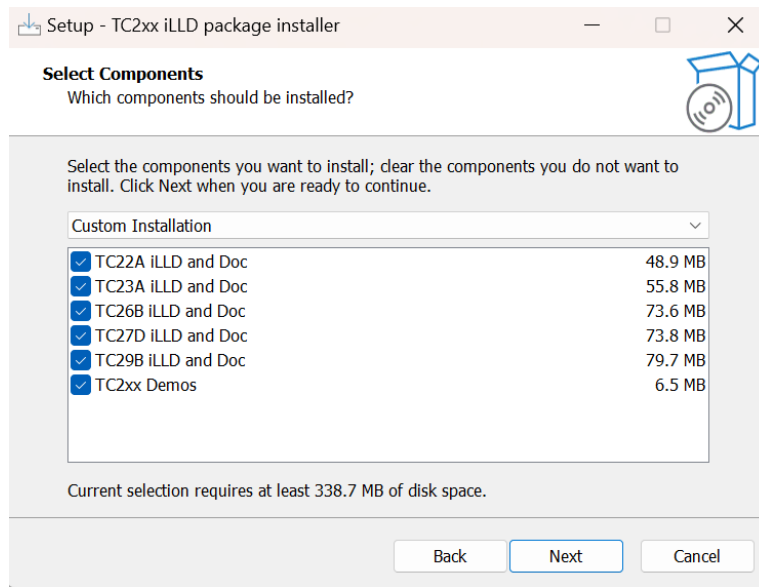
Note: If you have the destination folder already, content will be overwritten.

- Select full installation if you want to select all the components.

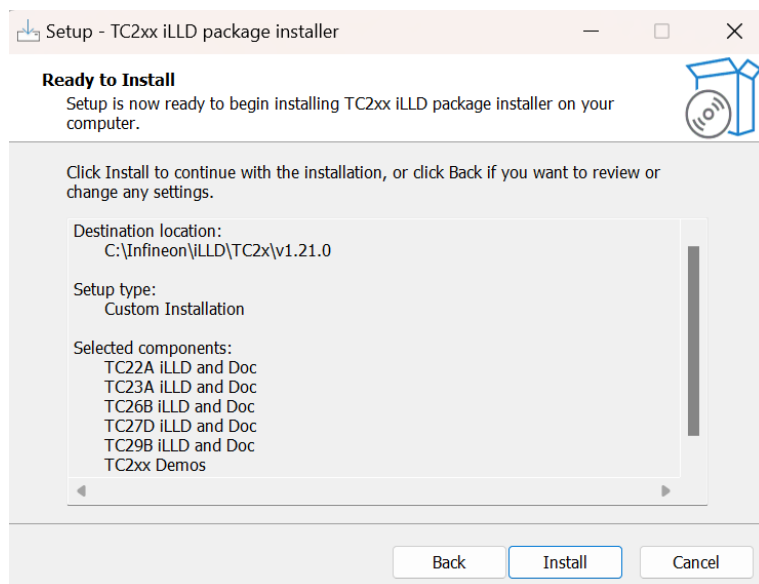


Else select only the Custom Installation to select the components you would like to install. You have selection possibility for individual derivate.

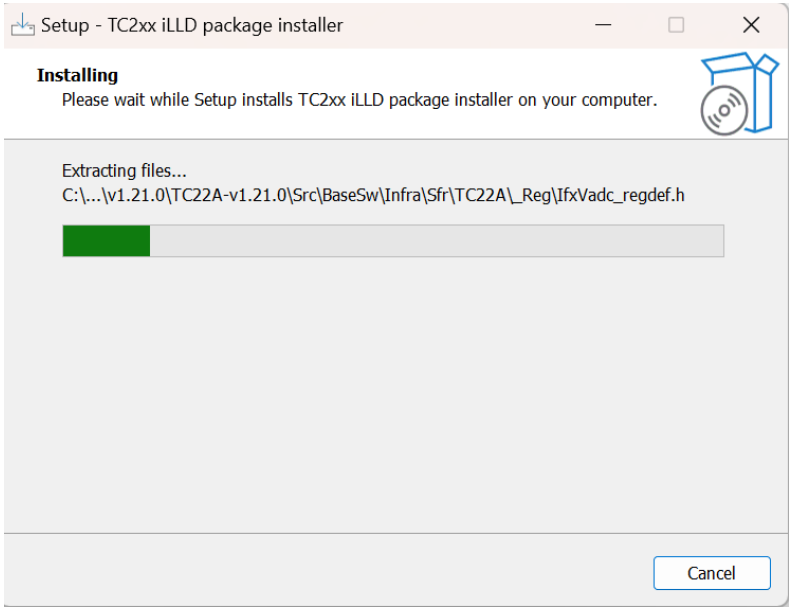
This page looks as shown below:



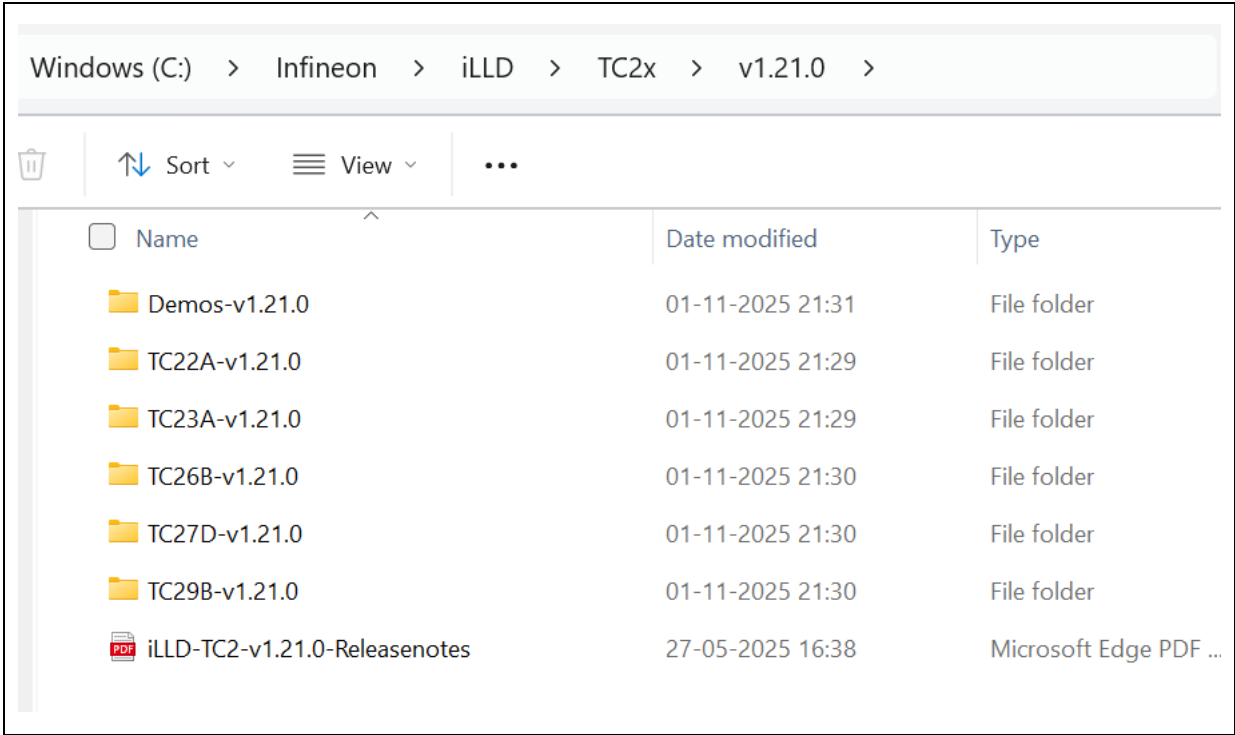
Click on **Next** button to go to **Ready to Install** page as below



Click the button “**Install**” to start copy process. This results in the following page



The package will be installed in C:\Infineon\iLLD\TC2x\<version>



7 Support packages

The Aurix 1G demo files are given for Evaluation Purpose only.

It contains the `Demos-v1.21.0` folder.

Table 11 Release zip contents

Package content	Description
<code>Demos-v1.21.0</code>	Demo Source files for TC2xx derivative

The application start-up software configuration files should be taken from the respective released Base Projects.

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